MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code**: MG5DSEBOT304, MG5DSEBBT302**
Course Title: **ANALYTICAL TECHNIQUES IN PLANT SCIENCE**

Duration: 2 Hrs **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A:
Very Short Answer Questions
Answer 10 questions
Each question carries 2 mark

	Questions	Taxonomy	CO
1	What is meant by maceration of plant tissue?	Remember	1
2	Define fixation in microtechnique.	Remember	1
3	Mention the composition of FAA.	Remember	1
4	Name any two vital stains used in plant microtechnique.	Remember	1
5	What is the role of a mounting medium in slide preparation?	Understand	1
6	State one use of Scanning Electron Microscopy in Botany.	Remember	3
7	Define spectrophotometer.	Remember	2
8	What is meant by UV-visible spectroscopy?	Remember	3
9	Expand FTIR.	Remember	2
10	Define lyophilization.	Remember	2
11	Define Rf value in chromatography.	Remember	4
12	Define electrophoretic mobility.	Remember	2

(2 x 10 = 20)

Part B:
Short Answer Questions
Answer 8 Questions
Each question carries 5 marks

		Taxonomy	CO
13	Explain the purpose and procedure of single and double staining in plant microtechnique.	Understand	1
14	Differentiate between rotary and sledge microtomes.	Understand	2
15	Describe squash and smear preparation	Understand	1
16	Outline the procedure for adjusting the pH of a solution using a pH meter, and state its principle.	Apply	5
17	State Beer-Lambert's Law. Describe the principle of photometric analysis.	Understand	2
18	Explain the procedure for enumeration of cells using a haemocytometer.	Apply	4
19	Explain the principle of centrifugation and enumerate the types of centrifuges.	Understand	2

20	Explain the procedure of thin layer chromatography (TLC).	Understand	3
21	Describe the applications of HPLC in plant science.	Understand	3
22	Explain the working and applications of SDS-PAGE.	Understand	3

(5 x 8 = 40)

Part C:

Essay Questions

Answer any one question

Question carries 10 marks

		Taxonomy	**CO**
23	Describe the steps involved in the preparation of permanent slides.	Understand	1
24	Explain the basic principle and working mechanism of light microscopy and summarise the applications of light microscopy in plant science research	Understand	2,3

(10 x 1=10)